Data-driven recommendations for increasing reliability in measures of infant word recognition

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Abstract

Children's early word recognition is both an important phenomenon in its own right and a key index of early language more broadly. The looking-while-listening paradigm (LWL), in which eye movements to the referent of a word are tracked to reveal online language comprehension, is a key paradigm for measuring word recognition in young children. Yet the analytic and design choices vary widely, and the consequences of these choices have rarely been investigated at a large scale. Here we use data from Peekbank, a repository of LWL data from studies with almost 4000 children (6 months – 5 years), to search for analytic and design decisions that maximize the reliability and validity of LWL measures. Based on these analyses, we recommend the use of long analytic windows for computing accuracy, and we do not find benefits for correcting for baseline preferences or excluding data from children or trials with low data. In our dataset, this combination of analytic choices increases reliability by approximately .20 and validity by .07. We also recommend that LWL paradigms be designed with a minimum of 10 target trials per child per condition to improve reliability and that experimentalists match their stimuli for animacy — image saliency differences due to pairing animate and inanimate images can mask word knowledge in infants. We hope that these recommendations lead to improved and unified standards for the measurement of children's word recognition.

Keywords: looking while listening, word recognition, language development, eyetracking, reliability, validity

Word count: X

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Introduction

Over the first few months of life, children become experts in their native language, organizing the phonological system of the speech that they hear and parsing speech into meaningful units. A bit later, children begin to show early signs of comprehension of words or phrases through simple behaviors such as turning their head when their name is called (e.g., “Hey, Billy!”) or reaching for an object that is named (e.g., “Where’s the ball?”). These everyday accomplishments mark important milestones in children’s development. Yet such global assessments may miss continuous processes of growth in underlying skills and the millisecond-level processes that take place during real-time language comprehension. Children, like adults, process speech incrementally, as it unfolds in time (Fernald, Pinto, Swingley, Weinberg, & McRoberts, 1998; Frank et al., 2025; Snedeker, 2013), and use features of the speech signal to extract meaning that will support language comprehension (Law & Edwards, 2015; Weisleder & Fernald, 2013).

These moment-to-moment processes are difficult to document without child-friendly procedures that allow precision in capturing how children’s behaviors are time-locked to speech. In particular, the “looking while listening” (LWL) procedure has emerged as a *de facto* standard for measuring language in infants and children, especially when they are younger than two years old, as other options for direct assessment during this period are limited (Frank, Braginsky, Yurovsky, & Marchman, 2021).[1] In the standard version of LWL, children see two pictures on a screen and hear a referring expression, e.g. “look at the ball”; their time-locked eye-movements to the ball over the distractor image are then taken to be evidence for language comprehension (Fernald et al., 2008).

LWL has been a key tool for enhancing our understanding of infants’ language processing and knowledge (Golinkoff, Ma, Song, & Hirsh-Pasek, 2013). For example, using LWL methods, we can measure increases in speed and accuracy in responding to familiar words (Fernald, Perfors, & Marchman, 2006; Frank et al., 2025) and increases in the detail of their phonological representations (Moore & Bergelson, 2024; Swingley & Aslin, 2000) during the first years after birth. Moreover, measures of children’s skill in the LWL task have also proven useful as an index of individual differences. For example, children who show more efficient language processing at 18 months show larger vocabulary size and more rapid vocabulary growth over the second and third years than children with less efficient processing (Fernald & Marchman, 2012; Fernald et al., 2006; Frank et al., 2025; Peter et al., 2019). Language processing efficiency has also been shown to link to later language, cognitive, and non-verbal outcomes in both typically-developing (Law & Edwards, 2015; Peter et al., 2019) and preterm children (Virginia A. Marchman, Adams, Loi, Fernald, & Feldman, 2016; V. A. Marchman, Dale, & Fenson, 2023; Virginia A. Marchman et al., 2018).

[1] Here we adopt LWL as a convenient label while recognizing that the procedure goes by a number of different names in the literature including the “intermodal preferential looking procedure” (Fernald, Zangl, Portillo, & Marchman, 2008; Hirsh-Pasek & Golinkoff, 1996; Reznick, 1990).

One key reason for the success of this research is the care that has been invested in the development of the looking-while-listening method. Past work has articulated best practices in collecting and analyzing looking-while-listening measures, with a particular focus on stimulus and design choices that make looking preferences maximally informative about word recognition processes (Fernald et al., 2008). However, despite widespread use of the looking-while-listening method in language development research, there is still limited consensus on how to best process and analyze time-series data of this type [from either adults or infants; Mirman, Dixon, and Magnuson (2008);Bergelson and Swingley (2012);Peelle and Van Engen (2021);Von Holzen and Mani (2012);Weaver, Saffran, and Arunachalam (2026)].

In the developmental field especially, this lack of consensus likely stems from structural factors: infant recruitment is time- and cost-intensive and datasets collected by individual labs are often small (Bergmann et al., 2018; Frank, Braginsky, Yurovsky, & Marchman, 2017). These issues make it difficult for any one researcher to systematically evaluate the consequences of specific analytic choices independently of the data collected for a particular study (cf. Peelle & Van Engen, 2021). Instead, researchers are often forced to make ad-hoc decisions or to adopt idiosyncratic lab practices that themselves originate from subjective judgments of what “seems to work”. By decreasing uniformity in the literature, this variation also increases the challenge of triangulating what precisely is being measured by LWL (Weaver et al., 2026). How can we put the measurement and analysis of infant word recognition and online language processing on firmer empirical footing?

Peekbank is a new data resource that offers a potential solution to this core issue by providing a large database of infant eye-tracking data from the looking-while-listening paradigm for exploring potential methodological questions (Zettersten et al., 2022). The most recent release of Peekbank at time of writing (version 2026.1), currently houses 44 datasets from infants across a wide age range [6 months - 5 years] and across a broad range of test words and materials, resulting in observations from over 3915 infants participating in the looking-while-listening task. The goal of the current manuscript is to use these data to systematically explore key analytic decisions that researchers make when using this paradigm, offering best-practices guidance for measurement.

Our approach throughout is to seek analytic choices that distill the best possible measures from the rich time course of behavior in the LWL task. Weaver et al. (2026) highlight open questions about both the reliability and validity of LWL. Starting with reliability, we aim for analyses that yield the maximal signal about individual children relative to both measurement error and irrelevant artifacts (Byers-Heinlein, Bergmann, & Savalei, 2022; Rhemtulla & Bottesini, 2022; Schreiner et al., 2024). Here, to measure reliability, we use both within-dataset inter-item reliability (ICC) and test-retest reliability within datasets with multiple LWL sessions close in time for the same children.

Turning to validity, we want to make analytic decisions that maximize the relationship with other measures that are thought to reflect related constructs (Flake & Fried, 2020; Kominsky, Lucca, Thomas, Frank, & Hamlin, 2022). Here, we select as our primary measure the correlations between LWL measures and parent reports of vocabulary measured with the MacArthur-Bates Communicative Development Inventories [MB-CDI, a popular parent report measure of early vocabulary that is widely used in conjunction with LWL tasks; Frank et al. (2021); Fenson et al. (1994)]. We focus specifically on the relationships between aggregate measures of vocabulary size from the MB-CDI and LWL data, given both the documented relationships between these (reviewed in V. A. Marchman et al., 2023) and the mixed evidence on item-level alignment between LWL and MB-CDI data (López Pérez, Moore, Sander-Montant, & Byers-Heinlein, 2024; Weaver & Saffran, 2026). This correspondence reflects the general relationship between word recognition in the moment (indexed by LWL) and the growth of early vocabulary over time (indexed by the MB-CDI) (Frank et al., 2025).

Our focus in this paper is on providing guidance for navigating some of the many decisions facing an investigator analyzing LWL data. Recent work using an earlier version of Peekbank (as well as other datasets) has investigated the psychometric properties of a range of different dependent measures in LWL tasks and their general construct structure (Sander-Montant, Fibla, Byers-Heinlein, & Killam, 2025). We focus instead on the consequences of varying analytic choices for two main measures, which tend to have the strongest psychometric properties (Sander-Montant et al., 2025). The first is *accuracy*: how much of the time a child looks at the target image compared to looking at either the target or distractor image over some period of time after the auditory stimulus. Children's responses unfold over time, but it is not known what start and end time maximizes the signal of children's response to the stimulus (Peelle & Van Engen, 2021). The second measure we focus on is *reaction time* – how long it takes a child to change where they are looking in response to the stimulus (Fernald et al., 2008). There are even more analytic options available for this metric – whether to measure to a child leaving the distractor or looking at the target, whether to measure RT on a log or raw scale, and how long after the start of the stimulus to start counting the RTs. For both of these measures, there are also questions about whether and when to exclude either children or trials.

While the LWL task is used for a wide variety of purposes, including measurements of individual differences (e.g., Peter et al., 2019) as well as measurements of experimentally-induced condition differences (e.g., Swingley & Aslin, 2000), here we focus only on what we refer to as "vanilla" word recognition items. These are items that test vocabulary directly with familiar targets, familiar distractors, and no informative carrier phrases or experimental manipulations. We focus on these items both because they are the most commonly used LWL items and because they are the case in which reliability and validity is most clearly defined. Vanilla trials are expected to reflect the construct of word knowledge and correlate with other measures of word knowledge, including vocabulary size.

The plan of the paper is as follows. We begin by overviewing the subset of the Peekbank database we selected for our analyses. We then make a set of targeted recommendations, providing evidence for each based on the impact of our recommendation on the average reliability and validity of the resulting measures. Our first set of recommendations focus on data processing decisions and our second set of recommendations concern design considerations. To foreshadow our results, we find that while many common data analysis practices provide relatively reasonable levels of reliability and validity, it may be possible to further optimize reliability and validity through careful, data-driven decision-making.

Methods: Dataset and approach

Peekbank is an aggregation of datasets from a number of LWL studies. This work is based on Peekbank version 2026.1 (released February 2026) which has 134929 trials from 3915 children and 44 datasets.

Inclusion criteria

We focus our exploration of data processing choices on a set of items where we have the clearest expectations around what we are trying to measure. While LWL is also used to study a variety of experimental effects, such as the effects of noise, mispronunciations, novel word learning, and expectations of mutual exclusivity, these items may index different constructs. We limit our analyses to what we term “vanilla” trials: trials that represent standard LWL trials testing familiar words without additional experimental manipulations. For a trial to be considered “vanilla”, both the target and the distractor images must be of familiar objects. For the auditory stimulus, the target word must be a real word that occurs once and is the first and only source of information about the target. The carrier phrase must be a well-formed, grammatical phrase, without any switches in language, and without any intentionally added noise or audio filtering. Vanilla trials could come from studies with only vanilla trials, from the filler or baseline trials of experimental studies, or even from studies where the experimental conditions met the vanilla criteria. (See SI: Criteria for Inclusion for more detailed criteria used to handle borderline situations.) While still heterogeneous, these “vanilla” trials have a shared interpretation for how looking behavior is expected to relate to underlying language knowledge and processing. We call the start of the target word in the audio the stimulus onset and label this as the zero point for timing. We only include trials from children 60 months or younger (see Figure S1 for age distributions). We do not place restrictions on the language of the trials or children’s language background.

Descriptives

This paper is based on 36 datasets, comprising 3171 children, 4851 administrations, and 75636 trials. There are MB-CDI comprehension scores for 632 children from 12 of the datasets, and MB-CDI production scores for 1143 children from 20 of the datasets. Our test-retest data comes from 5 datasets with multiple administrations within 1.5 months of each other, covering 365 children (788 administration pairs). See Table 1 for a list of included datasets. Figure S2 has time course plots showing the number of trials and timing of available data for each dataset.

Table 1
Descriptives of the included datasets.

	Dataset Name	Pct Trials	Trials	Admins	Subjects	Age	CDI	Test-retest	Language
1	Adams et al. (2018)	18.4%	13899	711	69	22 [13 - 38]	x	x	English
2	Fernald & Marchman (2012)	15.6%	11828	679	122	22 [17 - 32]	x	x	English
3	Weaver et al. (2024)	6.5%	4948	248	141	16 [14 - 24]		x	English
4	Fernald et al. (2013)	5.8%	4359	178	80	20 [17 - 26]	x	x	English
5	Hurtado et al. (2008)	5.3%	4035	136	76	21 [17 - 27]	x		Spanish
6	Fernald et al. (2006)	4.9%	3692	229	63	20 [15 - 25]	x		English
7	Yurovsky et al. (2013)	3.7%	2810	413	413	34 [12 - 60]			English
8	Yurovsky et al. (2017)	3.1%	2350	327	327	37 [8 - 60]			English
9	Bergelson & Swingley (2012)	2.8%	2081	84	84	12 [6 - 21]	x		English
10	Kartushina & Mayor (2019)	2.7%	2072	69	69	7 [6 - 9]	x		Norwegian
11	Weaver & Saffran (2026)	2.4%	1788	69	69	19 [18 - 20]	x		English
12	Weisleder & Fernald (2013)	2.3%	1769	58	29	22 [18 - 27]			Spanish
13	Yurovsky & Frank (2017)	2.3%	1747	285	285	26 [13 - 59]			English
14	Borovsky & Peters (2019)	2.2%	1676	79	79	18 [17 - 20]	x		English
15	Potter & Lew Williams (2023)	2.1%	1626	67	67	24 [21 - 27]	x		English
16	Yoon et al. (2015)	2.0%	1500	199	199	43 [13 - 60]			English
17	Sander-Montant et al. (2023)	1.8%	1381	123	123	22 [12 - 31]	x		English, French
18	Mahr et al. (2015)	1.6%	1232	29	29	21 [18 - 24]	x		English
19	Egger et al. (2020)	1.4%	1052	54	54	18 [18 - 19]	x		Dutch
20	Garrison et al. (2020)	1.4%	1046	35	35	15 [12 - 18]	x		English
21	Hurtado et al. (2007)	1.3%	1010	49	49	24 [15 - 37]			Spanish
22	Pomper & Saffran (2017)	1.0%	785	82	82	17 [14 - 19]			English
23	Frank et al. (2016)	1.0%	781	108	108	35 [12 - 60]			English
24	Bacon & Saffran (2022)	1.0%	754	38	38	23 [22 - 24]	x		English
25	Ronfard et al. (2022)	1.0%	737	40	40	20 [18 - 24]	x		English
26	Perry & Saffan (2017)	0.9%	718	42	42	20 [19 - 22]	x		English
27	Casillas et al. (2017)	0.9%	651	23	23	32 [9 - 48]			Tseltal
28	Swingley & Aslin (2002)	0.8%	596	50	50	15 [14 - 16]	x		English
29	Pomper & Saffran (2016)	0.6%	471	60	60	44 [41 - 47]			English
30	Kremin et al. (2023)	0.5%	410	60	60	42 [37 - 49]			English, French, Spanish
31	Pomper & Saffran (2015)	0.5%	389	25	25	40 [38 - 42]			English
32	Moore & Bergelson (2022)	0.5%	384	29	29	18 [16 - 20]	x		English
33	Potter et al. (2019)	0.4%	309	44	23	23 [18 - 29]	x	x	English
34	Byers-Heinlein et al. (2017)	0.4%	284	48	48	20 [19 - 21]	x		English, French
35	Pomper & Saffran (2019)	0.3%	249	44	44	40 [38 - 43]			English
36	Pomper & Saffran (2018)	0.3%	217	37	37	39 [38 - 43]			English
	Total	100%	75636	4851	3171	22 [6 - 60]	21	5	

Approach

Throughout we use two reliability metrics and two validity metrics. Reliability refers to how consistently measurements are measuring the same thing. One of our measures of reliability is an intra-class correlation (ICC). Specifically, we use a measure of inter-rater reliability that measures (within a dataset) how much different items (target labels) are consistent with each other in "rating" the subjects (kid-administrations). See SI: Details about ICC for more specifics. Another measure of reliability is test-retest reliability, which measures the correlation for measures on the same child at two different administrations within 1.5 months of each other. This measure is only possible on a subset of the datasets that closely spaced repeat administrations. While we are limited in how much data we have for test-retest reliability, this is the metric that is most important for measuring stable individual differences.

In addition to having reliable measures, we also care that our measurements correspond to the construct of interest (vocabulary and language processing speed). For validity, we correlate with CDI production measures and CDI comprehension measures, from the datasets where we have CDI administrations. We use both short and long form CDIs, using the proportion of items a child was marked as knowing as our measure. In general, correlating with production and comprehension in the expected directions is desirable, although we also may expect that LWL measures are detecting distinct aspects of language learning compared to what is represented in the CDI (see Frank et al., 2025 for a detailed analysis).

All the code and data to reproduce the analyses is available at https://github.com/peekbank/peekbank-method [TODO: anonymize for review if needed].

Data processing recommendations

Our first set of recommendations focuses on data processing decisions. Looking while listening data undergoes significant post-processing to go from the experiment to the data that is analysed. Experiment design, eye-tracking, and eye-gaze coding considerations are covered elsewhere (Fernald et al., 2008; Olson et al., n.d.; Venker et al., 2020; Wass, Smith, & Johnson, 2013), so we focus here on the process of going from a series of timepoints to the measures of accuracy and reaction time that are analysed. There are a number of decisions involved, including what windows and time cutoffs to use, and what exclusions to make around missing or low amounts of data.

Recommendation 1: Measure accuracy over a long window.

In LWL studies, accuracy is typically calculated as the fraction of time a child is looking at the target divided by the time they are looking at the target or the distractor, over a time window of interest. A key decision is therefore the time window of analysis. Typically windows start shortly after the point of disambiguation and continue until several seconds later; this window is intended to capture the peak of children's stimulus-driven looks to the target. However, in practice, studies vary in what windows are used, and often window selection is a locus for analytic flexibility (choosing windows post hoc to maximize a particular effect) (Peelle & Van Engen, 2021). Common analysis windows include starting 300 milliseconds after stimulus onset and ending 1800 ms after stimulus onset (following Fernald et al., 2008) and 367ms to 2000ms after stimulus onset (e.g., Swingley & Aslin, 2002), although windows that extend until 3 or 4 seconds after stimulus onset are also used (e.g., Bergelson & Swingley, 2012; Bergelson & Aslin, 2017; López Pérez et al., 2024). In this analysis, we begin from first principles and ask what windows maximize reliability and validity on word recognition trials.

Analytic approach. We first calculated the ICC reliability for accuracy across datasets, computing average ICC across a range of different window onsets and offsets. Next, we added the additional metrics of test-retest reliability and—to assess validity—CDI correlations, which were calculated by dataset for each combination of parameter values. To aggregate across datasets, we used a linear mixed-effects model with random intercepts by dataset, weighted by the number of administrations. We interpret the fixed-effect contrasts as what differences in outcome measure would be expected from switching from one set of parameters to another on a generic dataset. We report estimates and 95% confidence intervals. The best data-processing options might be different for children of different ages,

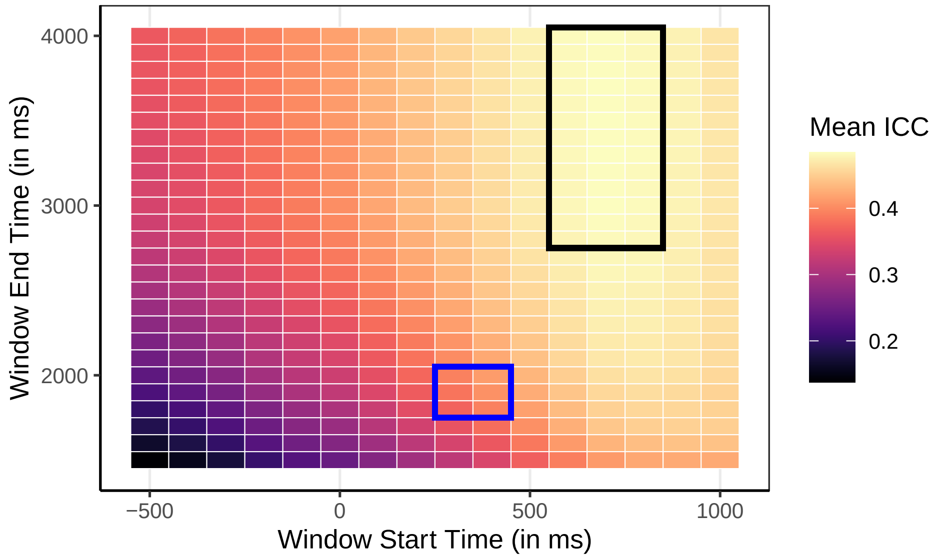

Figure 1. A heat map showing the average ICC across datasets for different start and end points for the accuracy window. A region with the highest ICC (best reliability) is outlined in black, and the region most commonly represented in the literature is outlined in blue.

so we ran the same linear mixed effect models separately for data just on children in each of 4 age groups: <18 months, 18-24 months, 24-36 months, and >36 months.

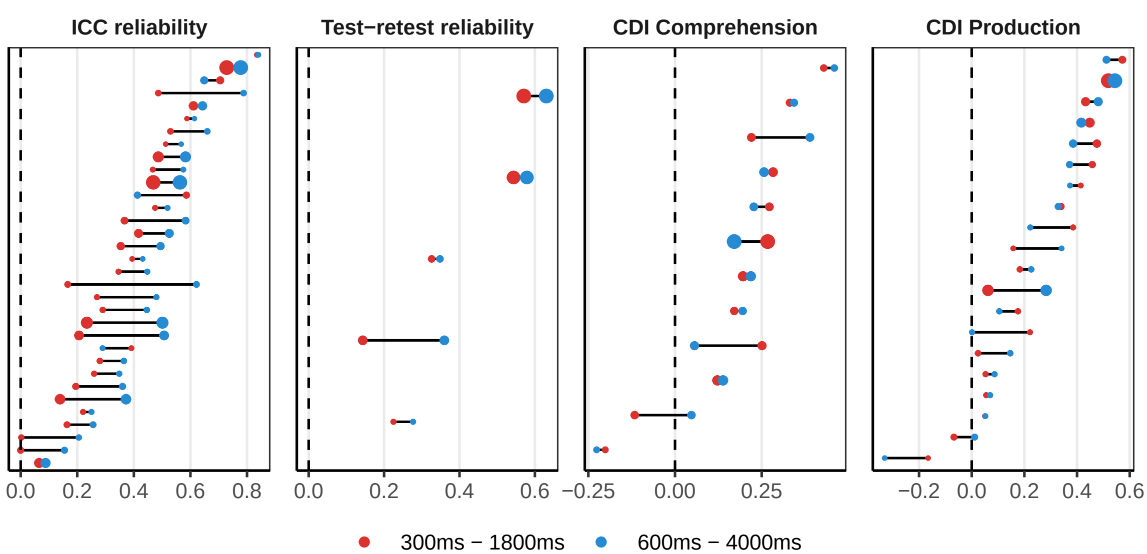

Figure 2. Per dataset outcome measures for two window options: 300ms-1800ms (red) and 600ms-4000ms (blue). Dot size is proportional within each panel to the number of administrations.

Results. Average ICCs across windows are shown in Figure 1. As expected, including time points from before the point of disambiguation decreases reliability. Our initial exploration found a region of high reliability with onsets in the 600-800 range and offsets in the 2800-4000 range, surrounded by a gradual decline in ICC reliability. A dataset by dataset comparison of the differences between a 300-1800ms window (representative of common practice) and a 600-4000 ms window are shown in Figure 2. Nearly all datasets have higher ICC reliability for the later and longer window than the shorter window. All 5 test-retest reliability datasets with close in time pairs of administrations also show increased reliability on this later and longer window. Our mixed effect model confirms that there are expected reliability gains from switching from a 300-1800ms window to a 600-4000ms one. ICC reliability is expected to increase by 0.12 [0.08, 0.16] and test-retest reliability by 0.07 [-0.02, 0.16]. There is no evidence of differences in validity between these two windows (-0.02 [-0.07, 0.03] for vocabulary comprehension or 0.03 [-0.02, 0.08] for vocabulary production).

These patterns generally hold across age. The overall level of reliability rises with age, as younger children generally fixate target images more consistently and experiments with younger children typically use fewer trials (Figure S3). However, the overall pattern of which time windows maximize reliability and validity at any given age is remarkably consistent across the different age groups (Figure S5). For instance, across different ages there is an improvement in ICC for using a 600-4000 window compared to a 300-1800ms one. For $<$18 months, the ICC difference is 0.06 [-0.03, 0.16]; for 18-24 months, 0.13 [0.07, 0.19]; for 24-36 months, 0.06 [0.01, 0.12]; and for 36+ months 0.15 [0.06, 0.23]. The main difference across ages is that when there is less consistency in accuracy measures generally (in the youngest age group), there is less improvement in reliability from longer window choices. For CDI production, there was no evidence for consistent age effects.

While we focused our analyses on the window that maximized reliability numerically (600 - 4000 ms), there are a wide range of window options with generally comparable reliability and validity measures (Figure S4). For instance, a 400 - 3000 ms window is not significantly different from a 600 - 4000 ms window on any of the outcome measures. Although we recommend starting between 400ms and 600 milliseconds to optimally balance validity and reliability, choosing when precisely the accuracy window starts is likely less important for maximizing reliability than choosing a long time window: regardless of window start time, increasing the window size consistently improves reliability (Figure 1).

Discussion. Overall, we find that LWL accuracy measures are reliable across a wide range of window choices; however, to maximize reliability, we recommend windows that start somewhere in the 400-600 ms range and extend to 3000-4000 ms (longer than the more canonical window ends of 1800 or 2000 ms). Windowing choices seem to primarily affect reliability measures and have limited effects on our validity measures (receptive and productive vocabulary). Why do later starting and later ending windows increase reliability? This suggests that, even as looking to the target often begins to decline in the post 2000ms window, looking behavior still contains reliable signal; at the same time, there may relatively little reliable signal in the early 300-600ms looking period. While some children may look quickly to the target and then look away, other children who look to the target later or who continue looking at the target at still contributing signal during later portions of a given trial. Overall, children in many datasets are in aggregate looking to the target more than the distractor during later looking periods (even if less than at the peak,

Figure S2).

Recommendation 2: Do not use baseline-corrected accuracy.

One commonly acknowledged issue in LWL studies is that looking behavior could be partly driven by differences in visual salience between the target and distractor images themselves (Fernald et al., 2008; Novack et al., 2025): Children may have a baseline preference to look at one image over another, even in the absence of auditory stimuli. Although thoughtful design choices can minimize these issues (see Recommendation 6), some analytic strategies also attempt to account for baseline preferences in dependent measures. While there are a variety of such strategies in the LWL literature (e.g., computing difference scores for specific target-distractor pairs; controlling for baseline preference), here we focus on one common practice: "baseline correction". In baseline correction, researchers use a difference measure between accuracy on a post-stimulus window and "accuracy" (looks to target versus looks to target or distractor) on a pre-stimulus window (e.g., Bergelson & Swingley, 2012; Weaver, Zettersten, & Saffran, 2024). The pre-stimulus window can be the 2 seconds prior to disambiguation (Weaver et al., 2024) or the full 3-4 second pre-stimulus period (Bergelson & Swingley, 2012).

Analytic approach. We considered baseline-corrected accuracies with various pre-stimulus windows, focusing on a a 600ms-4000ms main window (although the results generalize to other main windows). We restricted ourselves to datasets with at least some data from 2000ms or more before stimulus onset (see Figure S2 for which datasets have data when). We also computed a "no baseline" accuracy, based solely on accuracy within a 600-4000ms time window with no correction. Baseline windows started at -1000, -2000, -3000 and -4000ms and ended at stimulus onset.

Results. Baseline-corrected accuracy led to worse reliability and validity than accuracy. As an illustrative example, Figure 3 shows per-dataset results for accuracy compared to a baseline-corrected accuracy using a -2000 - 0 baseline window. This result held generally: All baseline window sizes that we considered led to worse reliability and validity measures compared to using normal accuracy without any correction. For a regular window of 600-4000, baseline correction is expected to decrease ICC by at least 0.1 [0.04, 0.16]; test-retest reliability by at least 0.19 [0.08, 0.3]; correlation with receptive vocabulary by at least 0.04 [-0.05, 0.12] and correlation with productive vocabulary by at least 0.09 [0.05, 0.13]. Similar patterns hold for different accuracy windows (Figure S6) and across different age groups (Figure S8).

Discussion. Why would incorporating baseline information lead to worse metrics? The literature on the psychometrics of difference scores (e.g., Lord, 1956; Rogosa & Willett, 1983; Trafimow, 2015) provides a valuable guide for evaluating the expected measurement properties of baseline correction. This literature points to two important features of looking while listening tasks: first, children's looking behavior is noisy from trial to trial, meaning that the reliability of proportion looking during both the baseline and the critical window is low; and second, the signal across baseline and critical portions of a trial is typically correlated (in part due to the very salience that the baseline correction is trying to remove).

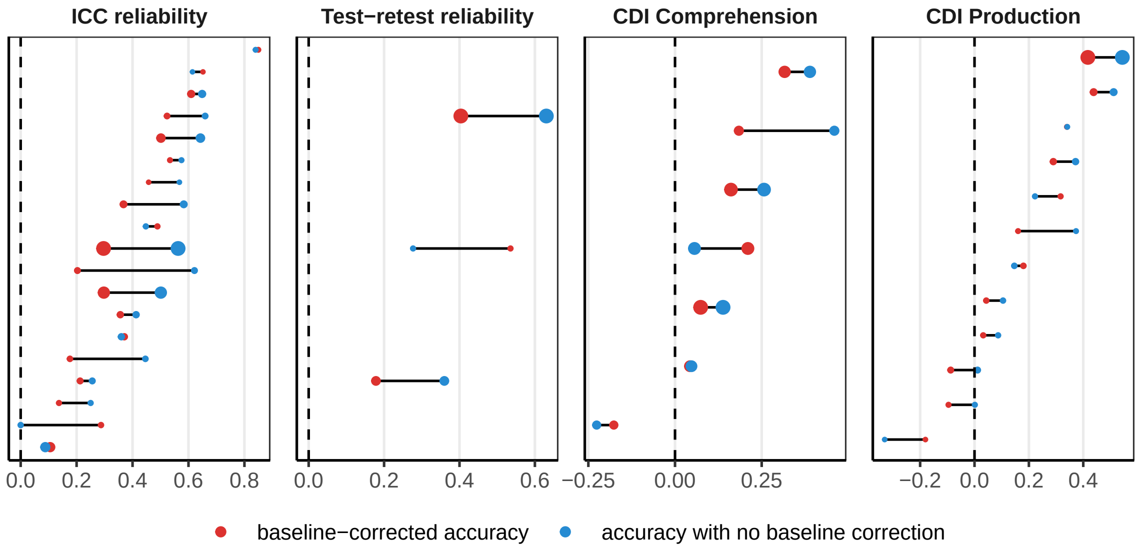

Figure 3. Comparison between a baseline-corrected accuracy in red and accuracy in blue across datasets and measures. Both use an accuracy window of 600 - 4000ms, the baseline window was -2000 - 0ms.

Unfortunately, it is under precisely these conditions – low reliability and high correlation between components – that difference scores exhibit the lowest reliability (Trafimow, 2015). Thus, while baseline-correction may lead to less biased grand average accuracy scores (by removing item salience), it also leads to less reliable and valid individual-level measures.

Recommendation 3: Measure land-based reaction times with a minimum reaction time of around 400 ms.

In addition to accuracy measures, LWL studies also provide the potential for computing children's reaction time [RT; Fernald et al. (2008)], i.e., the speed with which the process familiar words. RT is typically computed on trials when the child is initially looking at the distractor and shifts to the target after stimulus onset, to provide a measure of the speed with which infants respond to the familiar word by shifting their gaze to its target. RT computations include many choice points:

- **Launching vs. landing**: RTs can be computed based on when a child's eyes leave from the distractor (launch-based RT) or when a child's eyes arrive on the target image (land-based RT) (Fernald et al., 2008).
- **Exclusion of short and long RTs**: Short and long RTs are often excluded because they are considered unlikely to be generated in response to the stimulus (Fernald et al., 2008).
- **Linear vs. logarithmic analysis**: Reaction time data are often analyzed using a logarithmic transform (consistent with the idea that reaction times are log-normally distributed and that effects on them may be multiplicative rather than additive) (Csibra, Hernik, Mascaro, Tatone, & Lengyel, 2016; Frank et al., 2025).

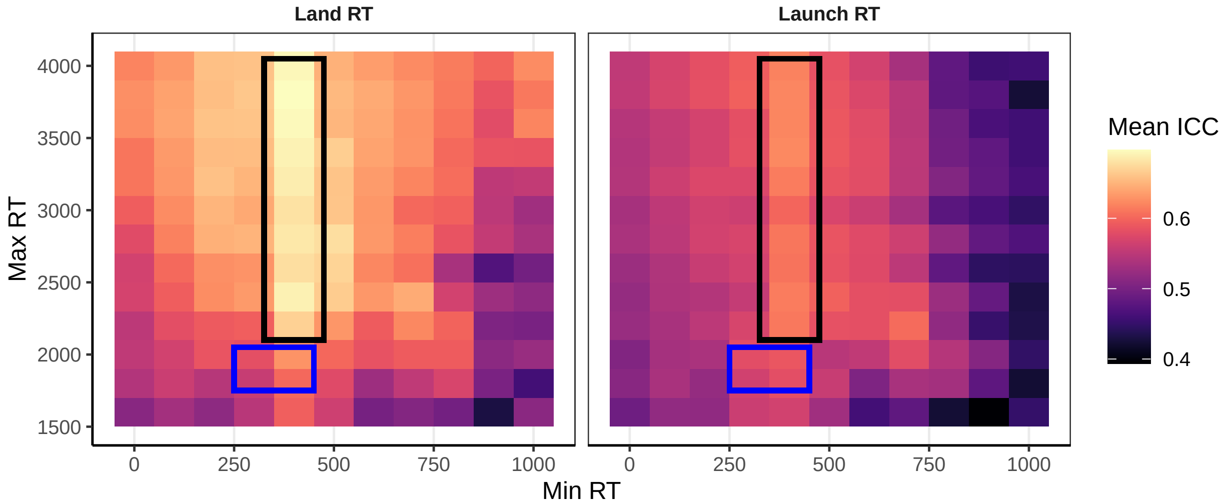

Figure 4. Heat map of different parameter options for measuring RT and how they on average affected ICC.

Analytic approach. We considered a wide range of options for measuring RT, fulling crossing a) launch- or land-based RTs, b) a minimum RT threshold for inclusion between 0 and 1000 ms, c) a maximum RT threshold for inclusion between 1600 and 4000, and d) raw or log-transformed RTs.

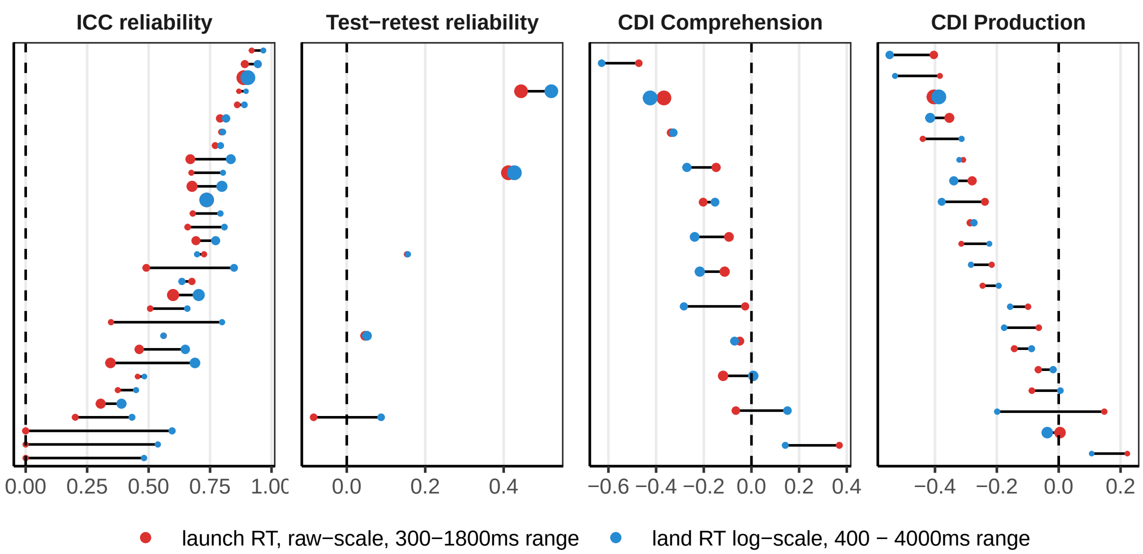

Figure 5. Comparison between two different sets of RT parameters. In red, an example of common practice, using a launch-based RT from 300-1800ms. In blue, an RT that maximizes ICC, with a land RT from 400 to 4000ms, on the log scale. Note that correlations with the CDI are generally expected to be negative (faster word recognition corresponds to more

Results. An initial exploration of ICC reliability (Figure 4) indicated that ICC reliability was maximized for land-based RTs; a minumum threshold for including RTs set at around 400 ms; and with higher maximum thresholds for RT inclusion. A comparison between a common set of parameters for calculating RT and an alternative that maximizes ICC are shown in Figure 5. Compared to computing the "standard" launch-based, raw RTs restricted to 300-1800ms, using land-based log RTs ranging between 400-4000ms is expected to increase ICC by 0.11 [-0.01, 0.23], with little change in test-retest reliability (0.04 [-0.18, 0.27]). These choices had little impact on CDI correlations (0.04 [-0.08, 0.17] for comprehension, and 0.03 [-0.06, 0.12] for production). Similar patterns hold for children of different ages (Figure S9 and Figure S11).

Discussion. Overall, RT measures complement accuracy measures and capture substantial signal, as indicated by the moderate ICC values across all choice parameters. A wide range of parameter options have similar properties (Figure S10 and Figure S12), but computing land-based RTs with an inclusion window of 400 - 4000ms maximizes reliability without negatively impacting validity. Note that the recommendation to compute land-based reaction times contradicts traditional approaches to reaction-time computation (Fernald et al., 2008) based on launch time; however, land-based RTs generally led to higher ICCs across our dataset. Log-scale transformations should be considered, but may depend on modelling and theoretical considerations.

Recommendation 4: Trial-level and session-level exclusions for low amounts of data are generally unnecessary.

A common issue in developmental research is when to apply trial- and session-level exclusions. For LWL studies, trial-level exclusions tend to be for contributing too few time points (e.g., due to tracker loss or looking away from the screen during the trial). Papers often use a combination of heuristic exclusions for inattention prior to coding looking behavior (Adams et al., 2018) and criterion-based exclusions trials based on the proportion of looking to the target and distractor during the trial. Cutoffs vary, with different experiments requiring minimum looking at the images of at least 20%, 50% or 66% of the critical window of a trial (Borovsky & Peters, 2019; Pomper & Saffran, 2019; Weaver et al., 2024). Studies also occasionally filter for looking at the target or distractor specifically at the point of disambiguation (C. E. Potter, Fourakis, Morin-Lessard, Byers-Heinlein, & Lew-Williams, 2019).

In addition to trial-level exclusions, studies also implement session-level (or child-level) exclusions based on the minimum number of valid trials in a sesson. For accuracy measures, children are often excluded if they had fewer than 25% or 50% of the total trials in a session, either within a condition, or overall (Adams et al., 2018; Garrison, Baudet, Breitfeld, Aberman, & Bergelson, 2020; Perry & Saffran, 2017). For RT, it is instead standard to require a minimum number of trials [e.g., two; Fernald et al. (2006); Adams et al. (2018)].

There are multiple possible ways that trial- and session-level exclusions could affect measurement properties. If trials with fewer valid looking points still provide signal of the child-level language behavior, excluding them will decrease reliability and validity. On the other hand, if these trials reflect noise and are inconsistent with behavior measured on trials with longer looking times, excluding them will increase reliability and validity. Similar possible outcomes apply to session-level exclusions: if children completing only a small number of trials per session are largely contributing estimates with increased noise, reliability and validity should increase when these sessions (or participants) are excluded; conversely, if these children are still contributing consistent signal, reliability and validity should decrease when session-level exclusions are applied.

Analytic approach. We consider 3 types of exclusions: trial-level exclusions for accuracy, administration-level exclusions for accuracy, and administration-level exclusions for RT (an administration in the Peekbank schema corresponds to one session for one child). We first explored trial-level accuracy exclusions, both on the basis of amount of off-target looks and on the basis of where the child was looking at target onset. For accuracy, we then considered per-session minimums either based on the number of valid trials, or based on the fraction of valid trials in an administration. The fraction of valid trials is the more common exclusion in the literature, but when drawing from many datasets that vary in the total number of trials, minimum trial count may represent a more comparable metric. For RT, we considered different cutoffs for the minimum number of valid trials required for inclusion.

We note that datasets contributed to Peekbank vary in whether excluded data was contributed, or whether only cleaned post-exclusion data was contributed. Thus, some datasets may not show effects of exclusions if those trials or administrations that would be excluded are missing from Peekbank.

Results. For exclusions, there are both considerations around reliability and validity, and around how much of the potentially useable data is being discarded. For trial-level exclusions, a requirement to look at the target and distractor at least 50% of the time results in 85.64 % (range: 27.22% - 98.92%) of trials being retained. These exclusions lead to an expected increase in ICC by 0.03 [0, 0.05] and no substantial change in test-retest reliability (-0.02 [-0.11, 0.08]). Differences in CDI correlation are not substantive (0.01 [-0.04, 0.06] for comprehension and 0 [-0.03, 0.03] for production).

Overall, there is a trade-off between reducing the amount of data and increasing the reliability of the remaining data. Filtering based on initial looking position, or having requirements for looking above .5 start to diminish the amount of data, and potentially negatively effect test-retest reliability and CDI measures (Figure S13 and Figure S14). Requirements for on-image looking rates between 0% (i.e., no exclusions) and 50% lead to globally similar outcomes numerically.

We then explored the effects of excluding sessions for too few trials. Here, we focus on the results for a minimum number of 8 total trials to illustrate general patterns in the results, but report outcomes for a range of proportion- and count-based thresholds in the supplement (Figures S15-S18). Only including sessions with at least 8 trials results in retaining 80.99 % (range: 0% - 100%) of total sessions. These exclusions show little evidence of increasing ICC (0 [-0.01, 0.02]) and or test-retest reliability (0.03 [-0.08, 0.14]) compared to retaining all sessions. Differences in CDI correlation are not substantive (0.01 [-0.04, 0.07] for comprehension and 0.01 [-0.03, 0.05] for production). We do not observe significant differences based on minimum trials below 10 or minimum fractions (Figures

S15-S18). Overall, excluding sessions with too few trials had minimal impact on reliability and validity.

We also considered the effects of minimum trial requirements on RT data. Only including sessions that contribute at least 2 trials results in retaining 90.93 % (range: 42.86% - 100%) of all sessions. Similar to previous results, these exclusions lead to a negligible change in ICC by -0.01 [-0.07, 0.04], a slight numerical increase in test-retest reliability by 0.03 [-0.08, 0.14], no change in CDI comprehension correlation(0 [-0.14, 0.15]), and a slight numerical increase in CDI production correlation (0.02 [-0.04, 0.08]). For RT, setting a low minimum threshold for number of trials (e.g., 2 trials) appears to lead to moderate improvements in test-retest reliability and slightly stronger correlations with CDI production.

Discussion. Overall, exclusions based on amount of data per-trial or per-session come with trade-offs between the amount of data and variability within the data. Standard practice within the field often errs on the side of excluding a large number of trials and sessions for missing data. Surprisingly, our results show that applying no exclusions or minimal exclusions often represents a reasonable choice, at least in terms of the psychometric properties of the resulting measurements.

Design recommendations

Our focus for this work was to explore how data processing choices affected the reliability and validity of the resultant measures. However, along the way, we identified several experimental design questions that had notable consequences for our central measures. We summarize two design recommendations that stood out as having the largest potential impact on reliability and validity (see supplement, section S8 for an additional analysis of order effects).

Recommendation 5: Design studies to have a minimum of 10 trials per child, per condition, or as many trials as possible for individual-difference studies.

Excluding children for low trial counts may increase test-retest reliability, but it also decreases the power and generalizability of results by excluding some children. Rather than retrospective exclusions, another approach is prospective: how many trials per child should one aim for to achieve adequate reliability and validity? Test-retest reliability is particularly important for studies that investigate longitudinal effects, as it assumes that the test sessions are representative of properties of the child at that age (Byers-Heinlein et al., 2022). To address this prospective question, we considered how reliability and validity measures would have changed if we had fewer trials per child. We filtered the data to include only children who had contributed at least N trials (for different values of N) and then repeatedly subsampled their trials and calculated our metrics on each subset.

The results for accuracy and RT are shown in Figure 6. ICC reliability is not sensitive to the number of trials per subject, so it is not shown. For accuracy, more trials per administration increases test-retest reliability, up to 20 trials (the highest we

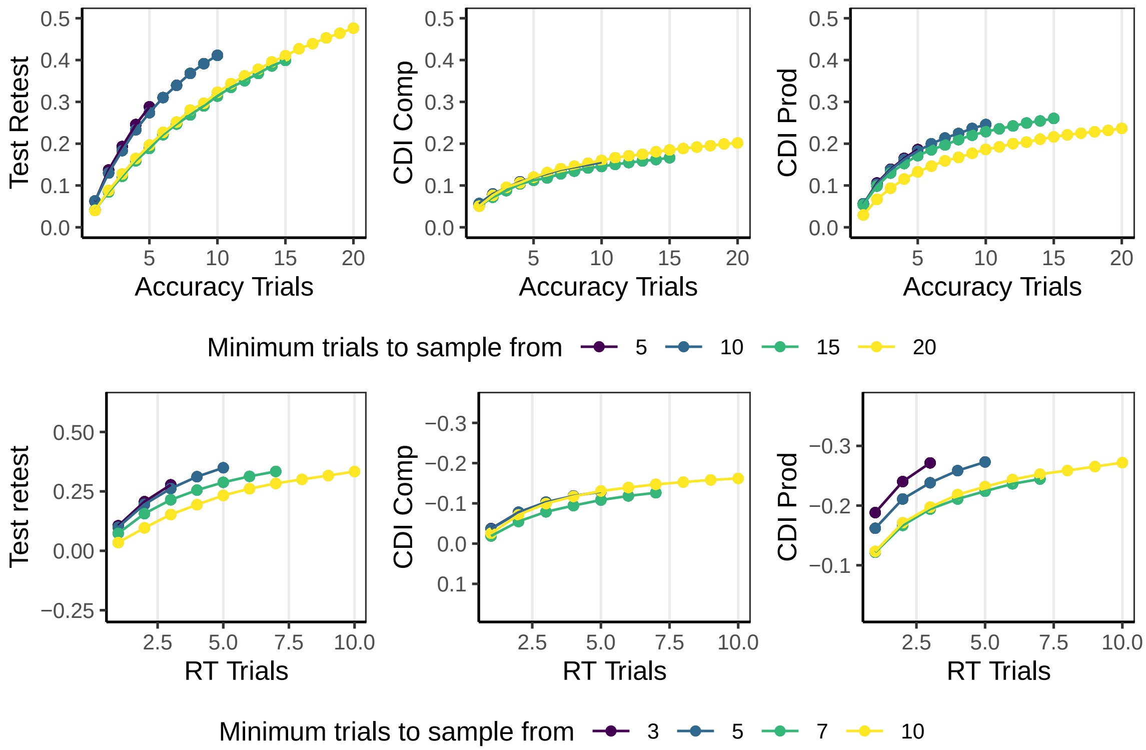

Figure 6. Simulations of measurement properties for different trial counts, created by down-sampling trials for all children with different trial minimums. Shown for accuracy window of 600 to 4000ms with no trial- or kid- level exclusions.

measured), although the gains per additional trial decrease with more trials. Validity with CDI measures also increased as trials increased, with smaller gains past 5 trials. Based on accuracy, we recommend at least 10 trials per condition in general, and as many as possible for individual differences studies where test-retest reliability is essential.

Similar considerations apply for RT, but at lower total trial numbers. Nearly every trial a child does results in a useable accuracy measure, but only around 1/3 of trials result in a valid RT measure in our data (in other cases, children start fixating the target or fixating neither). For test-retest reliability, having at least 5 RT measures per administration leads to noteable improvements; gains beyond 5 are hard to measure because of the low number of subjects with sufficient trials in multiple sessions, but more trials appears to be beneficial. For CDI measures, there are substantial gains for having 2 trials compared to 1 and continued benefits to including additional trials, up to the limit of trial numbers we are able to simulate based on available data. Overall, for measuring individual differences in processing speed, designing studies to include as many trials as possible appears to be central to adequate measurement.

Recommendation 6: Match target and distractor images based on animacy.

One concern with LWL trial design is that children may have preferences for which image to look at separate from the auditory stimuli, and some of their looking behavior may be independent of the auditory stimulus. One approach is to decrease the preference discrepancy between a pair of images, so that overall, in the absence of auditory cues, children prefer to look at both images approximately equally. While there are a number of factors that could play into children's image preferences (Fernald et al., 2008; Nothdurft, 2000; Pomper & Saffran, 2019), one factor that stood out in the data was animacy. Children have a strong preference for looking at animals and humans (and parts of humans). When trials have one animate and one inanimate, children look more at the animate in the pre-stimulus period (Figure 7) and are slower to respond when the inanimate is named. In contrast, for trials with target and distractor items matched on animacy, children's early looking patterns for animate and inanimate targets are highly similar. This difference is particularly relevant for the younger children, whose stimulus-contingent looking behavior is weakest, so more of the looking behavior is driven by animacy; however, effects of animacy were apparent in children up to the age of three (Figure S21).

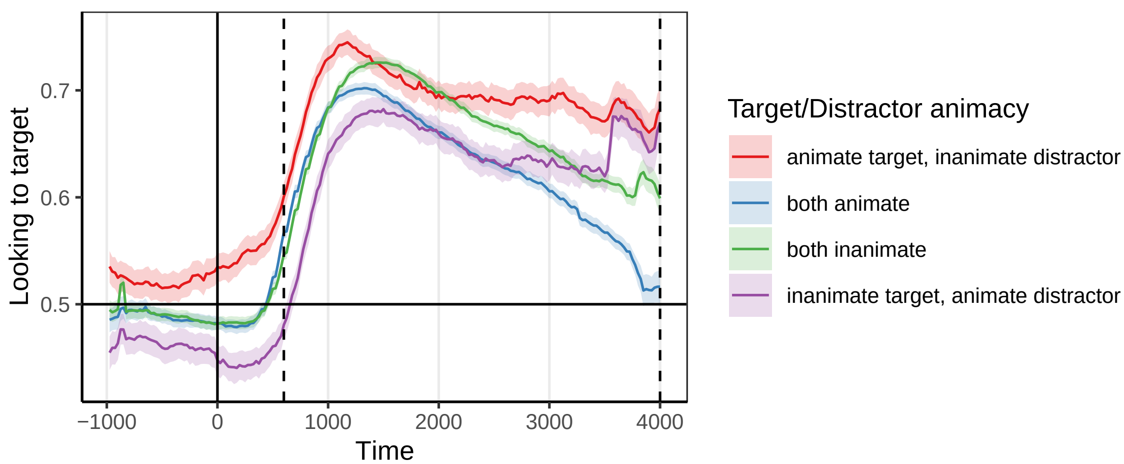

Figure 7. Looking time profiles for different combinations of target and distractor animacy.

How much can this matter?

To quantify the potential cumulative consequences of different combinations of choices, we computed the difference in reliability when making all of the recommended choices with respect to data processing to an instance in which a researcher made a set of unfortunate choices that are nevertheless consistent with the literature. Consider two researchers analyzing accuracy in the same LWL dataset, Norm and Rex. Researcher Norm makes a set of reasonable choices based on practices observed in other LWL papers, choosing to analyze a short analysis time window (300-1800 ms); apply baseline correction (baseline window: -2000-0 ms); exclude trials on which infants are not looking for at least

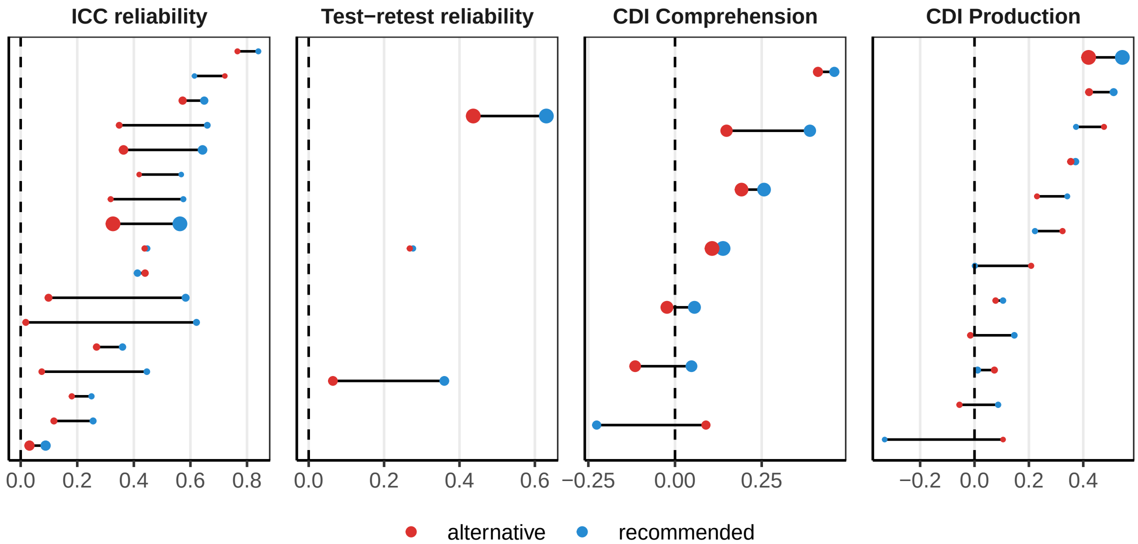

Figure 8. Differences in dependent measures between the recommended accuracy measurement and a sensible-sounding, but unfortunate alternative set of parameters.

60% of the critical window or not looking at the target onset for insufficient attentiveness; and require that infants contribute at least 50% of all trials for inclusion. Researcher Rex, on the other hand, analyzes the same data and instead chooses parameters based on the data-driven recommendations above: a critical window of 600-4000ms, no baseline correction, and including all trials and sessions. Researcher Rex would gain substantial increases in ICC (0.2 [0.13, 0.27]) and test-retest reliability (0.21 [0.13, 0.29]). Additionally, Rex could expect small increases in validity for correlations with comprehension (0.07 [-0.03, 0.17]) and production (0.07 [0, 0.13]). Despite both researchers making choices consistent with past practices, we would expect Rex to derive a measure with significantly improved psychometric properties compared to Norm, and therefore be in a substantially better position to study individual differences and longitudinal change.

General Discussion

Children's early word recognition is an important construct for the study of language learning, but the use of the LWL paradigm is hindered by uncertainty about data analysis. How can we maximize reliable, valid signal in the analysis of data from this paradigm? We took a practical, data-driven approach to this question, using a large-scale dataset of infant looking-while-listening data to quantify the impact of different analysis choices on core measurement properties, leading to six core recommendations for data processing and experimental design.

For data processing, our strongest take-homes from these analyses concern window choice and baseline correction. First, longer time windows (e.g., windows that end around 3000 ms post target word onset) maximize the reliability of accuracy measures, with no negative consequences for validity. While average accuracy tends to peak earlier in the timecourse of a trial (typically 1000-2000 ms post target word onset), leading shorter windows to be preferred in the literature, these findings suggest that later looking windows still capture signal that boosts the consistency of word recognition measures. Second, baseline correction, while an intuitive solution to imbalances in the saliency of images, causes large decrements in reliability; we therefore generally recommend avoiding this practice, especially when maximizing reliability is a high priority (e.g., when investigating individual differences). Finally, we formulate recommendations for choosing reaction time cutoffs (around 400ms tends to maximize reliability) and find that implementing trial- and child-level exclusions based on the amount of data contributed tend to have minimal consequences. Overall, choices about windows of analysis, approaches to exclusions, and how scores are computed can all have meaningful impacts on measure reliability.

One fascinating question that emerges from our analysis concerns the purity of the construct that LWL measures. In our analysis, long accuracy windows increased reliability more than they did validity. One speculative reason for this difference is that signal from later in the timecourse might not be as tightly coupled to language per se, and hence might instead reflect more general features of attention. Future research could use this difference as a starting point to further parcellate the attentional and linguistic contributors to language-driven eye-movements.

A more general take-home from our data processing analyses is that intuitive data-based practices, e.g., looking at patterns in grand-average time courses, may not always be a trustworthy guide to child- or item-level psychometric properties. Past recommendations for time window selection have sometimes been motivated by inspecting consistent patterns of looking in grand-averages, and, e.g., noting when looking begins to decline across participants. These heuristics — while based in important patterns in looking behavior — may not always lead to optimal choices with respect to core measurement properties such as reliability. Wherever possible, we recommend that measurement choices be guided by direct estimates of psychometric properties in order to maximize the signal in our measures of infant behavior (Byers-Heinlein et al., 2022; Sander-Montant et al., 2025; Weaver et al., 2026).

Among our design-focused recommendations, the most striking result is the impact of animacy on infant looking behavior. When an image of an animate referent was paired with an image of an inanimate referent, infant looking was systematically biased towards the animate image. This finding has important consequences for estimating word recognition: accuracy for animate targets is substantially overestimated and accuracy for inanimate targets becomes substantially underestimated when animacy is not balanced in the target-distractor pairing. The results are especially dramatic for younger infants (under 18 months): looking for familiar inanimate targets can appear to be at chance when paired with an animate distractor. In other words, an improper balancing of saliency could make an infant appear to not know a word, even though they do. Once animacy is balanced in the design, animate and inanimate targets have highly similar recognition curves across all ages. These results emphasize the importance of balancing saliency in image pairings as much as possible in looking-while-listening designs. This is especially true given that many of the current analytic approaches to correcting for saliency imbalances *post-hoc* rely on difference-based measures (baseline correction; item-pair correction) which decrease reliability.

Limitations & Generalizability

The scope of our recommendations is fundamentally limited by the characteristics of the Peekbank dataset. First, while the Peekbank database is by far the largest dataset of its kind, and one of the largest harmonized infant eyetracking datasets, it is not a comprehensive or representative set of looking-while-listening studies. To highlight just a few of these limitations, (1) the data is predominantly English word recognition collected from infants living in the U.S., and (2) the dataset is a convenience sample from labs who made their data openly available or otherwise accessible on request from the Peekbank team. Looking-while-listening studies as a whole are likely more heterogeneous than what is represented in Peekbank — the datasets considered here come predominantly from a small group of labs or networks of researchers with similar training in looking-while-listening data collection.

At the same time, there is also substantial heterogeneity in the studies included in Peekbank, due to many studies pursuing different research questions and therefore making a variety of distinct design choices (e.g., surrounding target words, image properties, and characteristics of the auditory stimuli). To mitigate this concern, our analyses focused exclusively on what we term "vanilla" trials, i.e., trials that present familiar words with familiar images with limited or no targeted manipulations of the stimuli. The main benefit of focusing on vanilla trials is that it allows for a clear interpretation of reliability and validity — trials are truly designed to measure word recognition, rather than some other aspect of language processing. We view the remaining heterogeneity of study properties (e.g., specific words; differences in age) as a strength of our approach — our findings generalize broadly across a wide range of items and ages. However, by focusing on vanilla trials only, we also limit our conclusions to measuring familiar word recognition; our findings do not address how data processing and design practices would generalize to other estimation goals, e.g., estimating the magnitude of differences between two experimental conditions (though there is good reason to think that some of the more consequential choices, e.g. baseline correction, are likely have a similar impact).

While we summarize the choices that typically maximize reliability, we note that we often see a range of choices with similar outcomes in our analyses. For example, in determining the best time window for computing accuracy, a choice of ending a time window at 2800 vs. 3000 or 3000 ms vs. 3200 ms leads to very similar outcomes. There is no one choice of time window that we can definitively infer as the "best" for maximizing reliability — instead, we see a pattern of graded improvements in reliability that follow a predictable pattern (longer time windows tend to increase reliability holding the onset of the accuracy window constant). While we sometimes articulate which choices lead to the highest reliability descriptively, these are at best approximations. We offer these windows as data-driven default choices (especially in cases where the decision would otherwise be ad-hoc).

Conclusions

Experimental design and data processing decisions can have profound impacts on the properties of infant measures. Our findings show how compiling large-scale datasets such as Peekbank can help us estimate the impact of a wide variety of experimenter decisions, generalizable across a wide range of ages, stimuli, and experimental contexts. More generally, establishing consistent practices and principles for reliable and valid measurement of infant looking sets the stage for asking central questions about how language skill develops.

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